Answer Key for Tutorial on Counting (Week 1)

Tier-1 Questions

1. (Chp6.1, Q3) See https://youtu.be/oyjV_00AreA
    a. 4^{10}
    b. 5^{10}

2. (Chp6.1, Q23) See https://youtu.be/27TIT9b-ioQ
    a. 128
    b. 450
    c. 9
    d. 675
    e. 450
    f. 450
    g. 225
    h. 75

3. (Chp6.1, Q29) See https://youtu.be/fu6DRm9d_HY
    52,457,600

4. (Chp6.1, Q45) See https://youtu.be/Y0ygybbsAdY
    60

5. (Chp6.1, Q47) See https://youtu.be/2994qKcnNxM
    a. 240
    b. 480
    c. 360

6. (Chp6.3, Q3) See https://youtu.be/uEweJl1OHjg
    720

7. (Chp6.3, Q13) See https://youtu.be/H9AQZlSx_g0
    $2(n!)^2$

8. (Chp6.3, Q19) See https://youtu.be/9jI_AJEpCLw
    a. 1024
    b. 45
    c. 176
    d. 252

9. (Chp6.3, Q23) See https://youtu.be/VbPK-Tve_lw
    609,638,400

10. (Chp6.3, Q27) See https://youtu.be/5t6w7F9bz5Q
    a. 12,650
    b. 303,600

11. Rounded answers are given below for reference, but you are expected to produce the formulae.
    See https://youtu.be/MkuqH3BFbZ0
    a. 1.87×10^{20}
    b. 1.02×10^{20}

12. Rounded answers are given below for reference, but you are expected to produce the formulae. See https://youtu.be/QyOt9tcrsB8
    a. 2.43×10^{18} 20!
    b. 1.32×10^{11} (10C5 x 10C5 x 4! x 5! / 2) x (5C5 x 5C5 x 4! x 5! / 2)

Tier-2 Questions

13. (Chp6.1, Q5) 42

14. (Chp6.1, Q24)
    a. 1000
    b. 4500
    c. 4536
    d. 6000
    e. 2829
    f. 6171
    g. 1543
    h. 257

15. (Chp6.1, Q30) $(26^3 \times 10^3) + (26^4 \times 10^2)$

16. (Chp6.1, Q44) ${}^{10}C_4 \times 3!$ or 1260

17. (Chp6.1, Q46)
    a. ${}^{9}C_5 \times 6!$
    b. ${}^{8}C_4 \times 6!$
    c. ${}^{8}C_5 \times 6! \times 2$

18. (Chp6.3, Q7) 15,120

19. (Chp6.3, Q15) 65,780

20. (Chp6.3, Q18)
    a. $2^8 = 256$
    b. ${}^{8}C_3$
    c. ${}^{8}C_3 + {}^{8}C_4 + {}^{8}C_5 + {}^{8}C_6 + {}^{8}C_7 + {}^{8}C_8$
    d. ${}^{8}C_4$

21. (Chp6.3, Q24) $10! \times {}^{11}C_6 \times 6!$

22. (Chp6.3, 33) ${}^{10}C_3 \times {}^{15}C_3 = 54{,}600$

23.
    a. $({}^{10}C_1 \times {}^{20}C_2) + ({}^{10}C_2 \times {}^{20}C_1) + ({}^{10}C_3 \times {}^{20}C_0) = 2920$
    b. 2.38×10^{15} (you are expected to produce the formula)

24.
    a. (i) 462, (ii) 25
    b. 1360800
    c. 2257920

~End